Open Dataset:

Leveraging Machine Learning for Smarter Restaurant Decisions

Anna Punty-Burkat | Andrea Mendoza

MSDS 422 Winter 2026 Final Project

Executive Summary

Our team analyzed **42K** restaurants and **2M** reviews (2018–2022) across 11 metro areas from the Yelp Open Dataset .

Developed 6 predictive models:

- ★ **2 text-based models from "Reviews" dataframe** (SVM + LSTM RNN) to predict ratings from review text.
- ★ **4 feature-based models from "Restaurant" dataframe** (2 Random Forest, LightGBM/XGBoost, DNN) using engineered features:
  - review_has_pos_words (dichotomous feature comprised of top 5 words most correlated with 5-star reviews)
  - avg_sentiment_polarity
  - avg_sentiment_subjectivity
  - subcategory_rating_score
  - city_mean_rating

Executive Summary (continued)

Key Insights:

- ★ **Text-based, review-level models:** Confirmed review language can predict ratings (accuracy: 96%), supporting future sentiment-based tools.
- ★ **Feature-based models:**
  - Top predictors include: text used, avg_sentiment_subjectivity (subjective language strongly influences ratings), Restaurant sub-category, sentiment polarity and average city rating
  - Least impactful: review_has_pos_words (generic positive terms less predictive than sentiment nuances), boolean features
  - X star ratings frequently miscategorized, biased towards higher ratings
  - Practical implications: ???

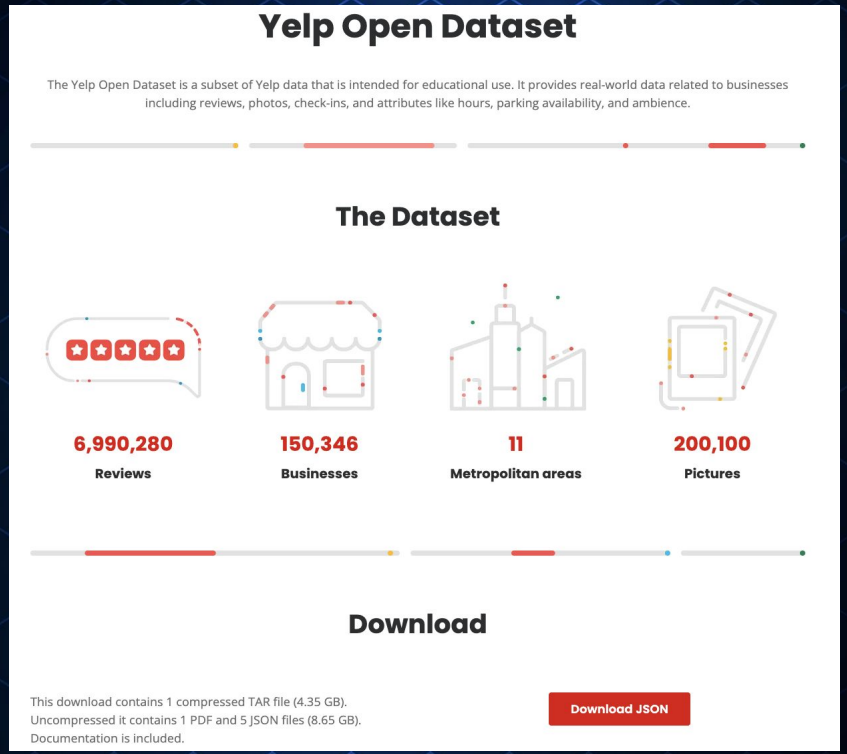
Business Problem + Objective (CRISP-DM Phase 1)

Can machine learning algorithms and Yelp review data help find patterns to support business success for large restaurant enterprises?

- ★ Businesses are heavily impacted by reviews on yelp. According to Harvard Magazine, "a one-star increase in Yelp rating leads to a 5-9 percent increase in revenue." For a Chicago Lou Malnati's location making 720 million/year, this translates to a minimum of 36 million/year.
- ★ The purpose of this project is to employ machine learning algorithms to train on Yelp review data, with the objective of uncovering patterns that lead to higher star ratings. While we used machine learning models on categorical data, we also used models on valuable text data, to try to predict star-rating based on words commonly used.

Data Understanding (CRISP-DM Phase 2)

- ★ Data source: Yelp Open Dataset, downloaded as JSON files.
- ★ Files used: Reviews + Business academic datasets.
- ★ Our project pulled Restaurant business in combination with reviews data:
  - Comprised of **42K** restaurants and **2M** reviews.



Data Understanding (CRISP-DM Phase 2)

- ★ The business dataset contained several columns with many missing values (such as Open 24 Hrs, Ages Allowed, Dietary Restrictions, Hair Specializes In, and Accepts Insurance), these values were used for testing, but were not effective.
- ★ Several businesses had null average star rating, which we dropped.
- ★ Final **Restaurant** dataset had **42K** restaurants.
- ★ Reviews data was loaded as object type, and did not have any nulls
    - There were few duplicates, which were dropped

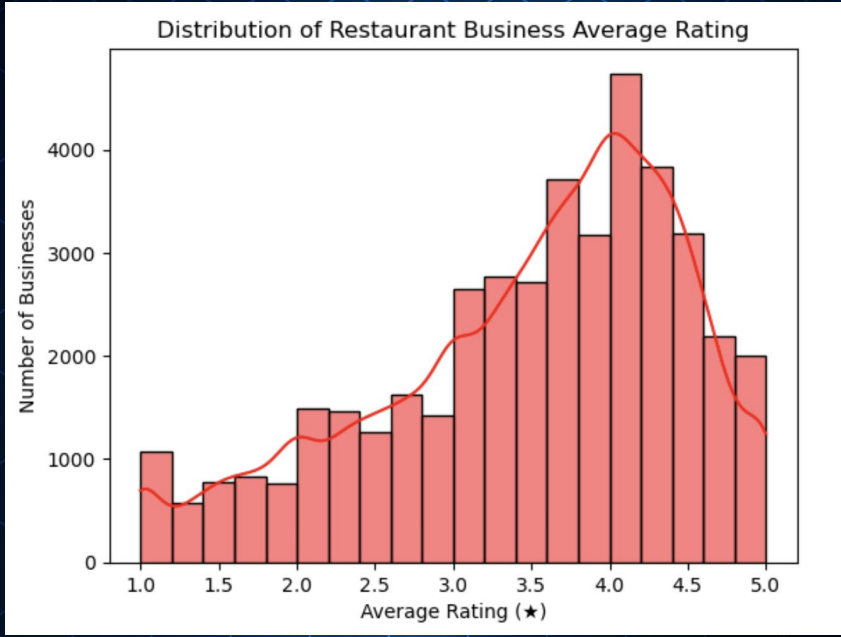
```
Columns with missing values (if any):
HairSpecializesIn              42274
AcceptsInsurance               42274
DietaryRestrictions            42274
Open24Hours                    42274
RestaurantsCounterService      42274
AgesAllowed                    42273
BYOBCorkage                    42251
Smoking                        42211
Corkage                        42210
BYOB                           42206
ByAppointmentOnly              42205
GoodForDancing                 42203
CoatCheck                      42198
BestNights                     42188
Music                          42173
BusinessAcceptsBitcoin         42161
DriveThru                      42144
DogsAllowed                    42049
WheelchairAccessible           42042
HappyHour                      42034
```

Data Constraints (CRISP-DM Phase 2)

- ★ High number of nulls in boolean fields (such as Open 24 Hrs, Ages Allowed, Dietary Restrictions, Hair Specializes In).
  - We wanted to use these types of features to categorize high and low ratings, their sparsity did not yield any strong results.
- ★ Data is only historical; Real time data is unavailable.
- ★ Large datasets presented computation processing challenges

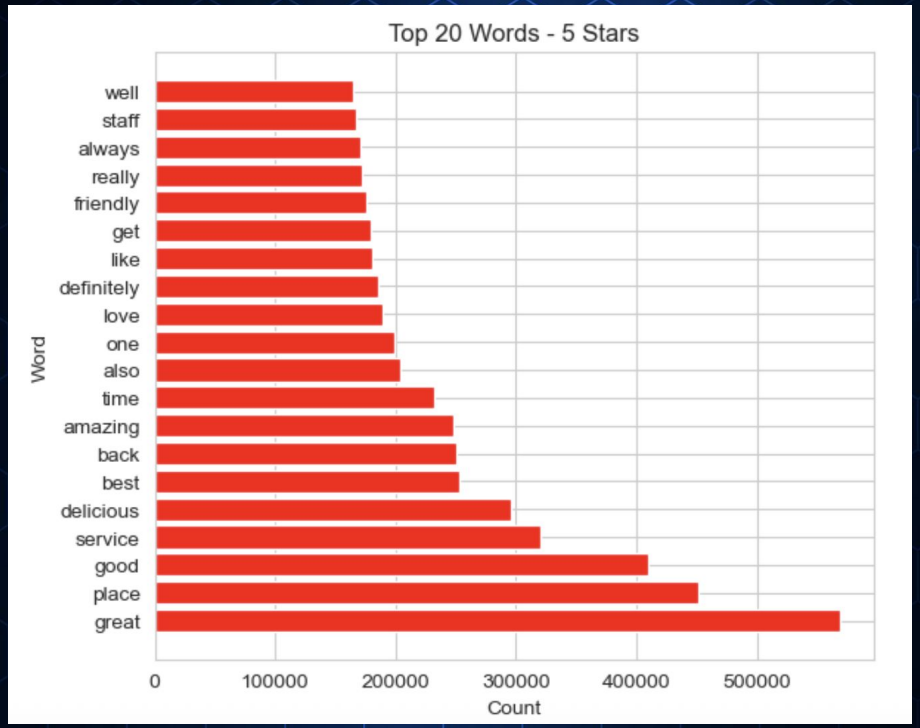
EDA Insights

- ★ Exploratory data analysis showed that the average star rating across all restaurants is 3.48 out of 5, with a standard deviation of 0.98.
- ★ Star ratings are negatively skewed, meaning that most restaurants had star ratings above 3.
- ★ Only 5% of the star ratings are greater than 4.80

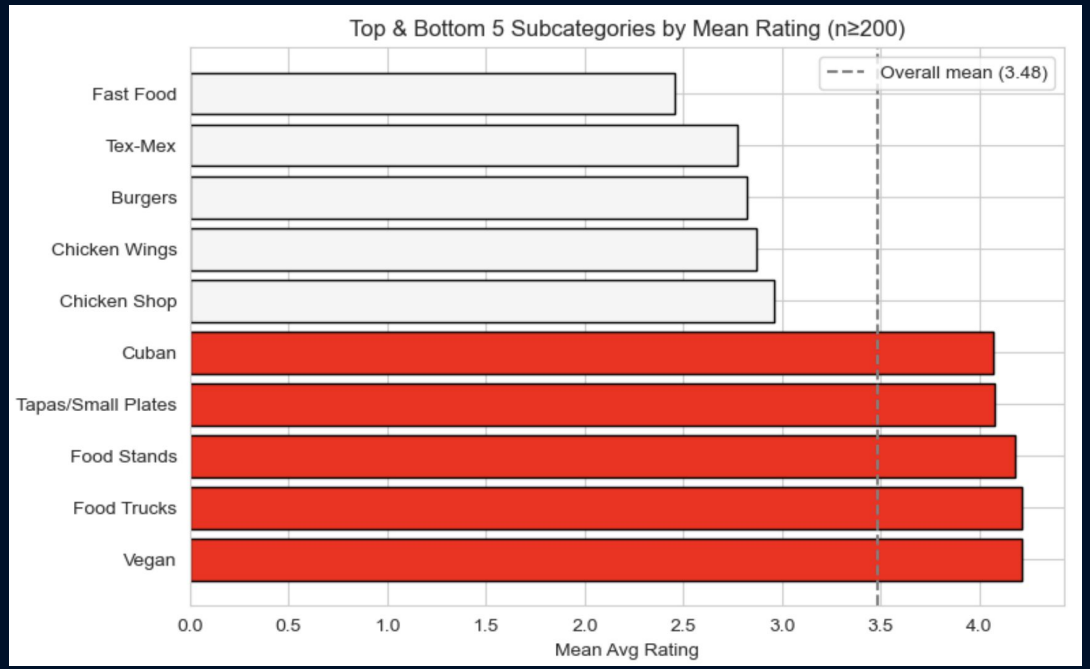


EDA Insights

- ★ Using the NLTK Stopwords Corpus, we counted the top words by star-rating.
- ★ Most used word in 5-star ratings is “great.”
- ★ Other unique words found in 5-star ratings include “amazing,” “best,” and “delicious.
- ★ Words highly correlated with 5 star ratings were used to create a dichotomous feature that flags a review if it includes a positive word.



EDA Insights



- ★ We correlated restaurant's subcategory types to observe star-rating variability.
- ★ **Fast food** had the lowest average star-rating
- ★ **Vegan a** nd food-truck businesses had the highest average star ratings
- ★ The overall mean was 3.48/5

Data Preparation (CRISP-DM Phase 3)

- ★ Selected years **2018-2022** of the review data and business ids matching those reviews.
- ★ Selected only **restaurant** business types
- ★ Split business type category into "Restaurant" and "Secondary business type," (restaurant types like fast food, cafes, etc.)

```
restaurant_df[['categories', 'secondary_category']].head()
```

	categories	secondary_category
0	Restaurants, Food, Bubble Tea, Coffee & Tea, B...	[Food, Bubble Tea, Coffee & Tea, Bakeries]
1	Burgers, Fast Food, Sandwiches, Food, Ice Crea...	[Burgers, Fast Food, Sandwiches, Food, Ice Cre...
2	Pubs, Restaurants, Italian, Bars, American (Tr...	[Pubs, Italian, Bars, American (Traditional), ...
3	Ice Cream & Frozen Yogurt, Fast Food, Burgers,...	[Ice Cream & Frozen Yogurt, Fast Food, Burgers...
4	Vietnamese, Food, Restaurants, Food Trucks	[Vietnamese, Food, Food Trucks]

- ★ Aggregated the review data by calculating the reviews count, average rating, and rating variability for each business
- ★ Merged summary features into main **Restaurant** dataset.

Methodology (CRISP-DM Phase 4): Tools + Hardware

Software + Tools

Language: Python

Data Processing: pandas, numpy, scipy

Text Processing: nltk, TextBlob, TfidfVectorizer, Tokenizer

Modeling:

- Traditional ML: scikit-learn (LinearSVC, RandomForest)
- Deep Learning: TensorFlow/Keras (LSTM, Dense)

Visualization: matplotlib, seaborn, folium

Utilities: tqdm (progress tracking), swifter (pandas optimization)

Hardware

Early EDA: Google Colab (GPU-enabled for LSTM/DNN)

Model Development: Local MacBooks (Anaconda/Jupyter Notebooks) + Google Colab

- Specs: Apple M2 + M4, 16GB RAM
- GPU Use: LSTM/DNN models used GPU in Colab; local runs used CPU

Methodology (CRISP-DM Phase 4): Model Selections

2 Predictive Text Analysis Models for Review Rating

Support Vector Machine (SVM):

- ★ Classifies restaurant ratings using patterns in review text
- ★ Uses TF-IDF to identify the most important words in reviews
- ★ Focuses on high vs. low ratings to create a clear decision boundary
- ★ Helps reveal language patterns associated with positive and negative experiences

Effective for high-dimensional text data and large vocabularies

Support Vector Machine (SVM):

- ★ Classifies restaurant ratings using patterns in review text
- ★ Uses TF-IDF to identify the most important words in reviews
- ★ Focuses on high vs. low ratings to create a clear decision boundary
- ★ Helps reveal language patterns associated with positive and negative experiences

Effective for high-dimensional text data and large vocabularies

Methodology (CRISP-DM Phase 4): Model Selections Continued

4 Predictive Models for Restaurant Rating Averages

Light Gradient Boosting Machine (LightGBM):

- ★ Processes categorical and continuous features in a single model
- ★ Efficiently handles mixed feature types and class imbalance
- ★ Computationally efficient for large-scale data
- ★ Provides feature importance rankings

Dense Neural Network (DNN):

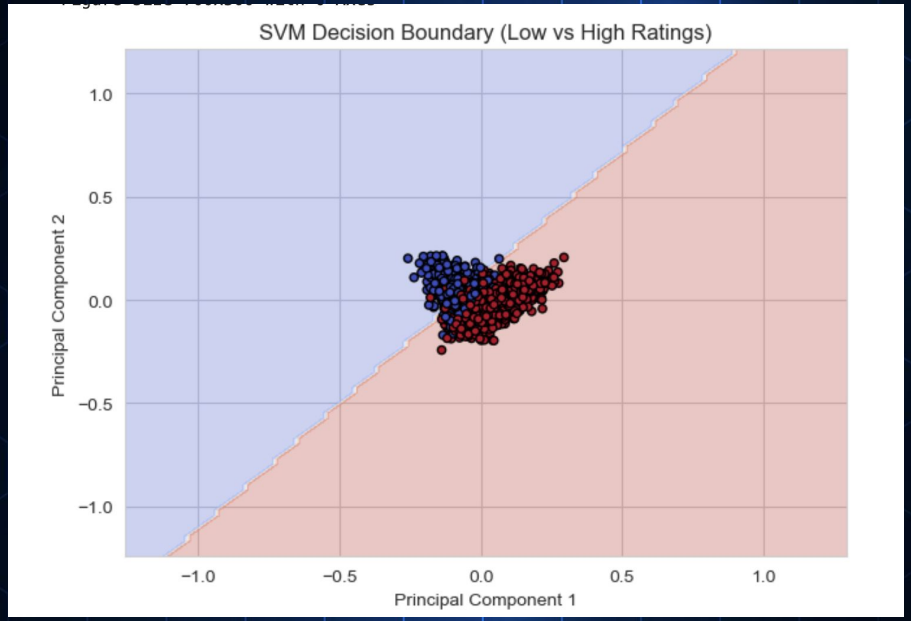
- ★ Captures complex, non-linear feature interactions missed by tree-based models
- ★ Uses normalized engineered features across multiple hidden layers
- ★ Finds relationships between features that simpler models can't detect.

2 Random Forest Models (Boolean features & Continuous Features):

- ★ Classifies restaurant ratings using boolean indicators and continuous variables
- ★ Captures non-linear relationships and complex feature interactions
- ★ Requires fewer statistical assumptions than traditional models
- ★ Used to assess whether restaurant characteristics and engineered features can distinguish high vs. low star ratings

SVM Model Build

- ★ Binary sentiment labels (high vs. low ratings) and converted the review text into TF-IDF features to train a model that learns patterns in word usage associated with higher and lower ratings;
- ★ Removed 3-star ratings from this binary categorization to reduce noise.



SVM Model Performance

The model was highly effective at classifying high vs low ★ reviews using word patterns

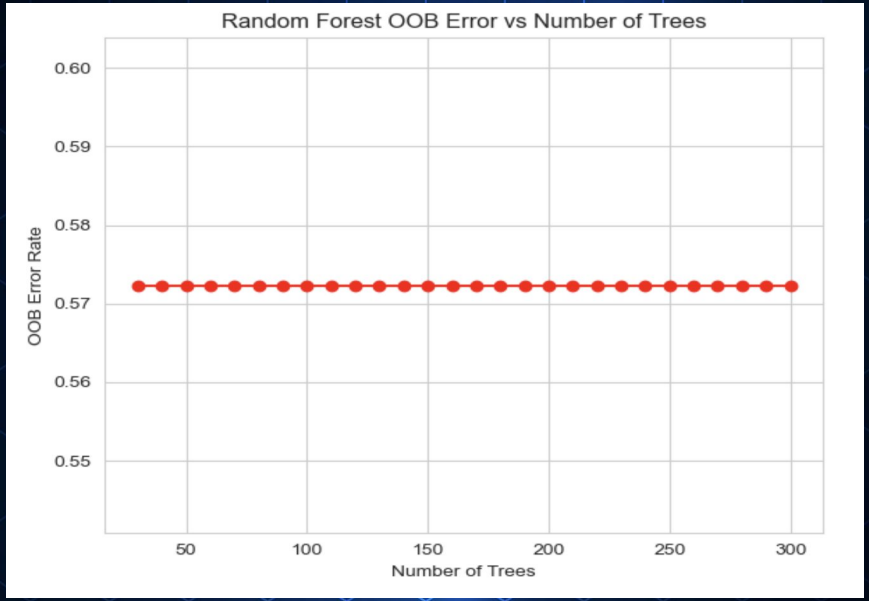
- **High★ ratings precision** = 99%, Recall = 96%, F1-score = 98%, indicating very strong identification of positive reviews
- **Low★ ratings precision** = 89%, Recall = 96%, showing the model correctly captured most negative reviews

```
Classification Report:
                    precision    recall  f1-score   support

 Low Rating (1-2)       0.89      0.96      0.93     86779
High Rating (4-5)       0.99      0.96      0.98    277435

          accuracy                           0.96    364214
         macro avg       0.94      0.96      0.95    364214
      weighted avg       0.96      0.96      0.96    364214


Confusion Matrix:
 [[ 83664   3115]
 [ 10396 267039]]
```

- **Confusion Matrix:**
  - 2.67K true positive high-star reviews
  - 3.5K misclassified low-star reviews

Random Forest Model - Boolean Features Build

- ★ Selected boolean features for binary target:
  - Low rating = 1–3 stars, High rating = 4–5 stars
- ★ Trained and evaluated a Random Forest classifier using stratified cross-validation
- ★ Assessed model stability with an out-of-bag (OOB) error curve
- ★ **Key finding:** Prediction error stabilizes after ~30 trees, indicating limited performance gains from adding more trees



Random Forest Model - Boolean Features Performance

- ★ **Test accuracy:** 43%, indicating weak overall performance
  - High-rating recall: 98% → model identifies most high-rating restaurants
  - Low-rating recall: 10% → model struggles to detect low-rating restaurants
- ★ The model is biased toward predicting high ratings and performs unevenly across classes
  - F1-score (high ratings): 0.56
  - F1-score (low ratings): 0.18

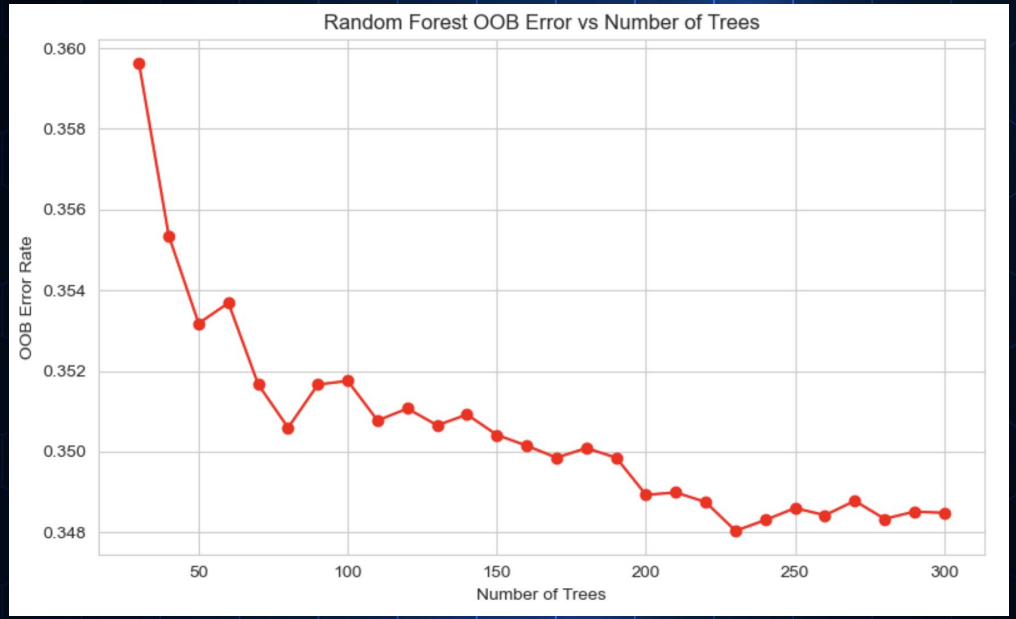
```
Test Accuracy: 0.4308

Classification Report:
                    precision    recall  f1-score   support

 Low Rating (<4)       0.88      0.10      0.18      5263
High Rating (≥4)       0.40      0.98      0.56      3192

        accuracy                           0.43      8455
       macro avg       0.64      0.54      0.37      8455
    weighted avg       0.70      0.43      0.32      8455


Confusion Matrix:
 [[ 521 4742]
 [  71 3121]]
```

Random Forest - Continuous Features Build

Used engineered features to predict rounded multi-class ratings:

- ★ Avg_sentiment_polarity
- ★ Avg_sentiment_subjectivity
- ★ Subcategory_rating_score
- ★ city_mean_rating

OOB error curve: Prediction error decreases as trees increase, stabilizing around ~200 trees, indicating improved model performance



Random Forest - Continuous Features

- **Test accuracy:** 65%
- Most accurately predicted 4★ ratings (78%),
  - Correctly predicted 2.6K 4★
  - confused many with 3★ (546) and five★ (563) ratings
- 2★ reviews were the next most accurately predicted class (62% precision and recall)

```
Test Accuracy: 0.6583

Classification Report:
              precision    recall  f1-score   support

           1       0.50      0.60      0.55       383
           2       0.62      0.62      0.62      1175
           3       0.61      0.64      0.62      1976
           4       0.78      0.70      0.74      3826
           5       0.51      0.60      0.55      1047

    accuracy                           0.66      8407
   macro avg       0.60      0.63      0.62      8407
weighted avg       0.67      0.66      0.66      8407


Confusion Matrix:
 [[ 231  136   13    1    2]
 [ 182  734  231   18   10]
 [  33  260 1267  384   32]
 [   6   36  546 2675  563]
 [   8    9   30  373  627]]
```

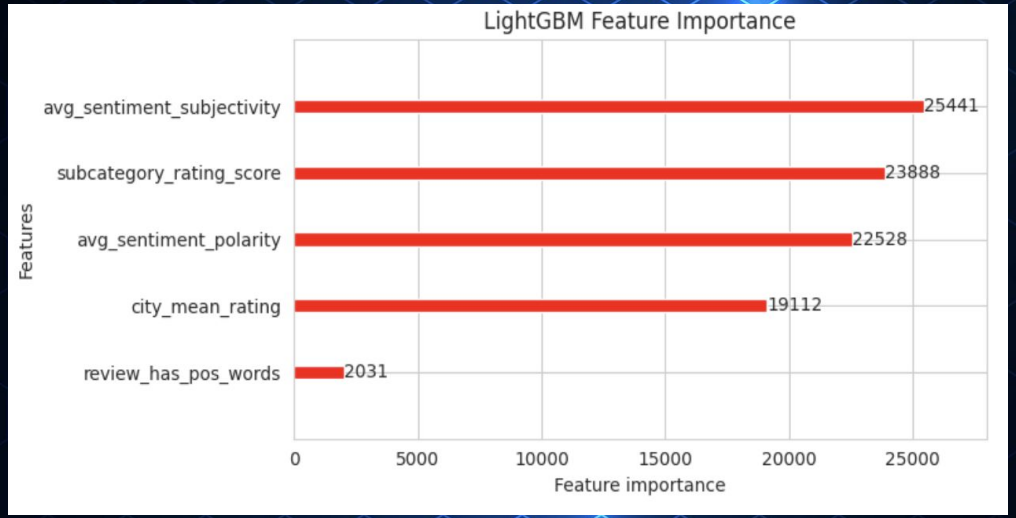
Light Gradient Boosting Machine (LightGBM) Build

Model Inputs: Trained on five engineered features from the business-level dataset to predict Average Rating (avg_rating):

- ★ 1 categorical boolean:
  - review_has_pos_words
- ★ 4 continuous variables:
  - Avg_sentiment_polarity
  - Avg_sentiment_subjectivity
  - Subcategory_rating_score
  - city_mean_rating

Feature importance ranking revealed:

- ★ avg_sentiment_polarity (most important)
- ★ subcategory_rating_score, city_mean_rating, and avg_sentiment_subjectivity (moderate importance, closer to polarity)
- ★ review_has_pos_words (least important)



Light Gradient Boosting Machine (LightGBM) Performance

```
Classification Report:
              precision    recall  f1-score   support

           1       0.49      0.64      0.56       383
           2       0.61      0.63      0.62      1175
           3       0.59      0.66      0.62      1976
           4       0.80      0.61      0.69      3826
           5       0.46      0.69      0.55      1047

    accuracy                           0.63      8407
   macro avg       0.59      0.65      0.61      8407
weighted avg       0.67      0.63      0.64      8407


Confusion Matrix:
 [[ 245  126    8    0    4]
 [ 194  746  207   15   13]
 [  38  291 1300  312   35]
 [  10   45  641 2320  810]
 [  11   11   30  270  725]]
```

Accuracy: 63.4% (CV) / 63.47% (test), indicating

Class performance: Best: 4-star ratings (F1=0.69, precision=0.80)

Worst: 1-star ratings (F1=0.56, recall=0.64)

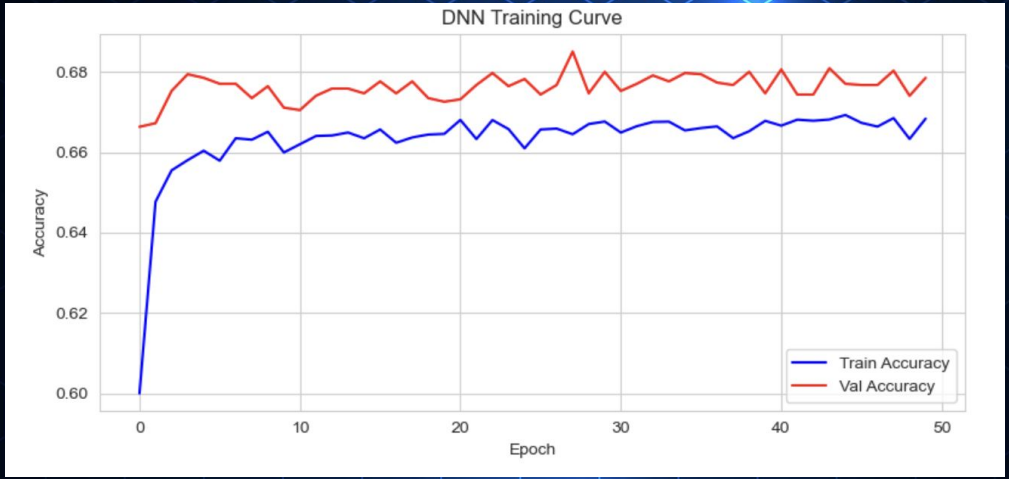
Common misclassifications:

- 5★ often predicted as 4★ (810 instances)
- 1★ confused with 2★ (126 instances)
- Trend: Higher recall for extreme ratings (1★/5★), but lower precision

Dense Neural Network (DNN) Build + Training

Model Inputs:

- ★ Trained on five engineered features (same as LightGBM) the business-level dataset to predict Average Rating (avg_rating)
- ★ 3-layer feedforward neural network with dropout regularization, designed for 5-class classification.



Training Curve: Converged/stabilized around epoch 5

- ★ Training beyond 10 epochs was unnecessary
- ★ Further hyperparameter tuning could reduce fluctuations

Dense Neural Network (DNN) Performance

```
Classification Report:
              precision    recall  f1-score   support

           1       0.69      0.27      0.39       383
           2       0.61      0.71      0.66      1175
           3       0.64      0.61      0.63      1976
           4       0.72      0.85      0.78      3826
           5       0.66      0.32      0.44      1047

    accuracy                           0.68      8407
   macro avg       0.67      0.55      0.58      8407
weighted avg       0.68      0.68      0.67      8407


Confusion Matrix:
 [[ 105  269    4    5    0]
 [  31  833  266   40    5]
 [  10  207 1208  545    6]
 [   2   35  389 3239  161]
 [   5   11   15  677  339]]
```

Accuracy: 68.09% (test), indicating

Class performance: Best: 4-star ratings (F1=0.78, precision=0.72)

Worst: 1-star ratings (F1=0.39, recall=0.27)

Common misclassifications:

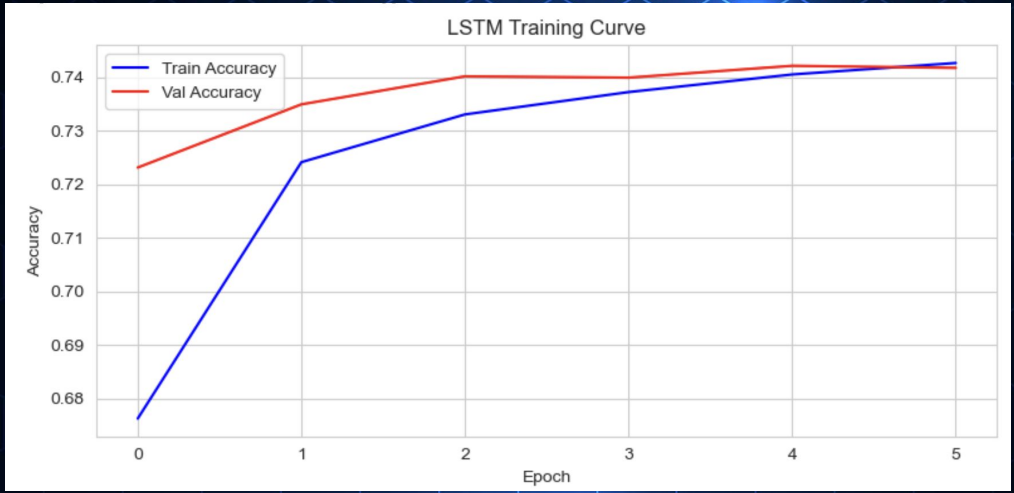
- 3★ often predicted as 4★ (545 instances)
- 1★ confused with 2★ (269 instances)

Trade Offs:

- High precision for 1★ (0.69) but low recall (missed 73% of actual 1★ reviews).
- High recall for 4★ (catches 85%) but lower precision (20% of 4★ predictions were wrong).

Long Short-Term Memory (LSTM) Recurrent Neural Network (RNN)

- ★ Trained LSTM recurrent neural network to predict star ratings from review text
  - Used one-hot encoding for 5-class classification
- ★ **Training curves:**
  - Validation accuracy was initially higher than training accuracy (good generalization); both later converged, indicating stable learning without clear overfitting



Long Short-Term Memory (LSTM) Recurrent Neural Network (RNN)

- **Accuracy:** 74%
  - The highest precision was in 5★ ratings (82%), with a recall of 93%.
  - Best precision score was for 1★ ratings (74%), and a recall of 85%.
- The model struggled to generalize 2–4 star ratings, frequently misclassifying them into other, especially higher, rating classes.

```
Test Accuracy: 0.7419

Classification Report:
              precision    recall  f1-score   support

           1       0.74      0.85      0.80     56140
           2       0.51      0.32      0.39     30639
           3       0.56      0.46      0.51     37190
           4       0.58      0.44      0.50     73774
           5       0.82      0.93      0.87    203661

    accuracy                           0.74    401404
   macro avg       0.64      0.60      0.61    401404
weighted avg       0.72      0.74      0.72    401404


Confusion Matrix:
 [[ 47953   4583   1746    446   1412]
 [ 11641   9663   6938   1327   1070]
 [  3322   4269  17183   9471   2945]
 [   766    314   4281  32641  35772]
 [   762     74    471  12012 190342]]
```

Model Evaluation Summary (CRISP-DM Phase 5)

Model	Use Case	Performance	Strengths / Weaknesses	Potential Next Steps
★ SVM ★	Predict review rating (high v. low)	**96%** Accuracy	Classifying complex and nonlinear text data / Binary Nature	Improving low ratings by including 3-star ratings to the low-rating class
Boolean Random Forest	Predict review rating (1-5)	**43%** Accuracy	Handles high-dimensional data/ nulls in data drove high misclassification	This model used variables with null data, which made it a weak model.
Continuous Random Forest	Predict review rating (1-5)	**65%** Accuracy	Better classification of 2-4 star reviews/ high misclassification	Reduce noise by rimplementic PCA
★ LightGMB ★	Predict avg. business rating	**63%** Accuracy	Feature importance extraction / low precision for extreme ratings	Fine-tuning hyperparameters + further feature engineering
Dense Neural Network	Predict avg. business rating	**68%** Accuracy	IDing non-linear trends, comprex relationships / low precision for extreme ratings	Tune architecture depth, further feature engineering, + custom loss functions
LSTM RNN	Predict review rating (1-5)	**74%** Accuracy	Long-term memory for sentence analysis/ bias towards 5-star ratings	

Model Deployment Strategy Recommendations for a Large Restaurant Enterprise (CRISP-DM Phase 6)

Based on analysis of 6 models using historical data from the Yelp Open Dataset (2018–2022), we recommend deploying two models in tandem to deliver real-time insights using live Yelp review data :

- ★ SVM for **predicting review ratings** in real time:
  - Highly efficient at classifying text in both high-rating and low-rating reviews

- ★ LightGBM for **forecasting business performance:**
  - Provides valuable feature importance identification (e.g., Avg_sentiment_subjectivity)

LightGBM Model Deployment Strategy Recommendations (CRISP-DM Phase 6)

Model Integration

- ★ Model should incorporate live Yelp review data
- ★ Deploy the trained LightGBM pipeline with engineered features to predict business average ratings.
- ★ Incorporate fine-tuned hyperparameters + further feature engineering

Real-time Prediction

- ★ Forecast subcategory and location-specific rating trends and identify underperforming restaurants.
- ★ Flag high-impact features (e.g., declining avg_sentiment_polarity) for managerial review.

Feedback Loop

- ★ Regularly monitor feature importance and model accuracy, retraining quarterly?? with new data
- ★ Use insights to improve operational strategies and feature engineering.
- ★ Consider: Integrating restaurant revenue data to further understand impact

SVM Model Deployment Strategy Recommendations (CRISP-DM Phase 6)

Model Integration

- ★ Model should incorporate live Yelp review data
- ★ Deploy the trained TF-IDF + LinearSVC pipeline
- ★ Incorporate PCA for dimensionality reduction and visualization of class separation

Real-time Prediction

- ★ Use the model to classify new incoming review text as low rating (1–2 stars) or high rating (4–5 stars).

Feedback Loop

- ★ Continuously monitor model performance
- ★ Retrain with updated review data
- ★ Use insights to support improvements in both restaurant operations and the model.

Lessons Learned + Conclusion

Lessons Learned

- ★ Model selection highly impact run-time,
- ★ Changing star-rating to a binary target variable increases model performance
- ★ Ensuing selection of features that have complete data for effective analysis
  - ○ Too much time spent on boolean features that include null values.

Conclusion

- ★ The combined use of **SVM** and **LightGBM** models, leveraging two datasets from the **Yelp** Open Dataset, enables large restaurant enterprises to use live Yelp review data to anticipate and mitigate the risk of low ratings and better understand high-review drivers, adjusting operational decisions accordingly.

Literature Review

- ★ O'Reilly Media. (2005). Python Fundamentals: Natural Language Processing (Lesson 11.03)
- ★ Yelp. (n.d.). Yelp Open Dataset. https://business.yelp.com/data/resources/open-dataset/
- ★ Harvard Magazine. (2011). Positive Yelp Reviews Lead to Increased Business. https://www.harvardmagazine.com/2011/10/hbs-study-finds-positive-yelp-reviews-lead-to-increased-business
- ★ Data Science Process Alliance. (n.d.). What is CRISP-DM? https://www.datascience-pm.com/crisp-dm-2/